MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MODEL QUESTION PAPER - THEORY STATISTICS
MG5DSCSTA302 - ANALYTICAL TOOLS FOR STATISTICS-I

Duration: 2 Hours **Maximum: 70 Marks**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer questions

Answer any **10** questions. Each question carries **3 marks**.

1. Find the supremum and infimum of the set $A = \left\{\frac{1}{5n};\ n \in I,\ n \neq 0\right\}$. [A] [1].
2. Prove that every sub set of a bounded set is bounded. [U] [1].
3. Is it true that 'a set of real numbers that is not open is closed'? Justify your answer [An][1].
4. Define subsequence of a sequence. Give example[U] [1].
5. Find the value of $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$[An][2].
6. State Leibniz's test for the convergence of an alternating series[A][2].
7. Evaluate $\lim_{x\to 0} \frac{\sqrt{4+x}-2}{x}$[An][3].
8. Distinguish between continuity and uniform continuity of a function [An][3].
9. State Darboux's theorem for derivatives [U] [4].
10. Give an example of a function that is continuous at a point but not differentiable there. [A] [4].
11. Discuss the differentiability of $f(x) = |x| + |x-1|$ at $x = 0$ and $x = 1$[An][4].
12. Verify Lagrange's mean value theorem for the function $f(x) = 2x^2 - 10x + 29$ on $[2,7]$ [An][4].

$10 \times 3 = 30$ marks

Part B

Short Essay questions

Answer any **4** questions. Each question carries **7 marks**

13. What are supremum and infimum of a set? Find the infimum and supremum of the set $A = \left\{\frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \ldots\right\}$ [An][1].
14. Prove that the sequence $\{S_n\}$, where $S_n = \frac{1}{n+1} + \frac{1}{n+2} + \ldots + \frac{1}{n+n} \quad \forall\ n \in N$ is convergent [An][1].

15. Test for convergence of the series $\frac{\alpha}{\beta}+\frac{1+\alpha}{1+\beta}+\frac{(1+\alpha)(2+\alpha)}{(1+\beta)(2+\beta)}+\cdots$ [A][2].

16. Show that the function $f(x)=x^2$ is uniformly continuous on $[-1,1]$[An][3].

17. Show that if $f_n(x)=\frac{n^2x}{1+n^4x^2}$, then $\{f_n\}$ converges non-uniformly on $[0,1]$.[An] [4].

18. Prove that if a function is differentiable at a point, then it is continuous at that point. Is the converse true? Justify your answer with a suitable example. [An] [4].

$4\times7=28$ marks

Essay questions

Answer any one question. Each question carries **12 marks**.

19. (a) State and prove Cauchy's Root Test for the convergence of a series of positive terms. [An] [2].

(b) Examine the series $\sum\left(\frac{n}{2n+1}\right)^n$ for convergence. [An] [2].

20. (a) Determine the constants and so that the function defined below is continuous everywhere:

$$f(x)=\begin{cases}1 & \text{for } x\le 3\\ ax+b & \text{for } 3<x<5\\ 7 & \text{for } x\ge 5\end{cases}$$ [An] [3].

(b) Discuss the continuity at $x=1,2$ and 3 of the following function:

$$f(x)=\begin{cases}x+4 & \text{if } x\le 1\\ x^2+3x+1 & \text{if } 1<x\le 2\\ 2x^2+x+1 & \text{if } 2<x<3\\ 8x-2 & \text{if } x\ge 3\end{cases}$$ [An] [3].

$1\times12=12$ marks